INVERTIS UNIVERSITY BAREILLY
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BAS-102 – Mathematics –I

MODULE-I

Matrices

Elementary row and column transformation, Rank of matrix, Linear dependence, Consistency of linear system of equations and their solution, Characteristic equation, Caley-Hamition theorem, Eigen values and eigen vectors, Diagonalisation, Complex and unitary matrices, Application of matrices to engineering problems.

MODULE-II

Differential Calculus-I

Leibnitz theorem, Partial differentiation, Eulers theorem, Curve tracing, Change of variables, Expansion of function of several variables.

Differential Calculus-II

Jacobian, approximation of errors, Extrema of functions of several variables, Lagranges method of multipliers (Simple applications).

MODULE-III

Fourier Series Periodic functions, Trignometric series, Fourier series of period 2 π, Eulers formulae, Functions having arbitrary period, Change of interval, Even and odd functions, Half range sine and cosine series.

Vector Calculus: Vector differentiation. Velocity, Acceleration of a particle moving on a space curve. Point function, Gradient, divergence and curl of a vector and their physical interpretations.

Text Books:-

1. H.K.Dass, Higher Engineering Mathmatics, S.ChandPublications.
2. B.S.Grewal, Engineering Mathematics, Khanna Publishers,2004.

Reference Books:-

1. R.K.Jain & S.R.K.Iyenger, Advance Engineering Mathematics, Narosa Publishing House, 2002.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna Publishers,2005.
3. E.Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons,2005.
4. C.Ray Wylie & Louis C. Barrett, Advanced Engineering Mathematics, Tata Mc Graw-Hill Publishing Company Ltd.2003
5. Peter V. O"Neil, Advanced Engineering Mathematics, Thomson (Cengage) Learning,2007.
5. Peter V. O"Neil, "Advanced Engineering Mathematics", Thomson (Cengage) Learning, 2007.